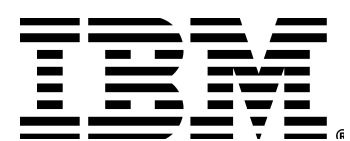


Containers & OpenShift Workshop

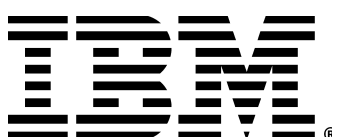
Alexander Mamani - IBM
Kate Queiroz - Red Hat
Christian Zaccaria - Red Hat
Winnie Imafidon - IBM



Let's do a recap of kubernetes



Red Hat



1. RECAP

a container is the smallest compute unit



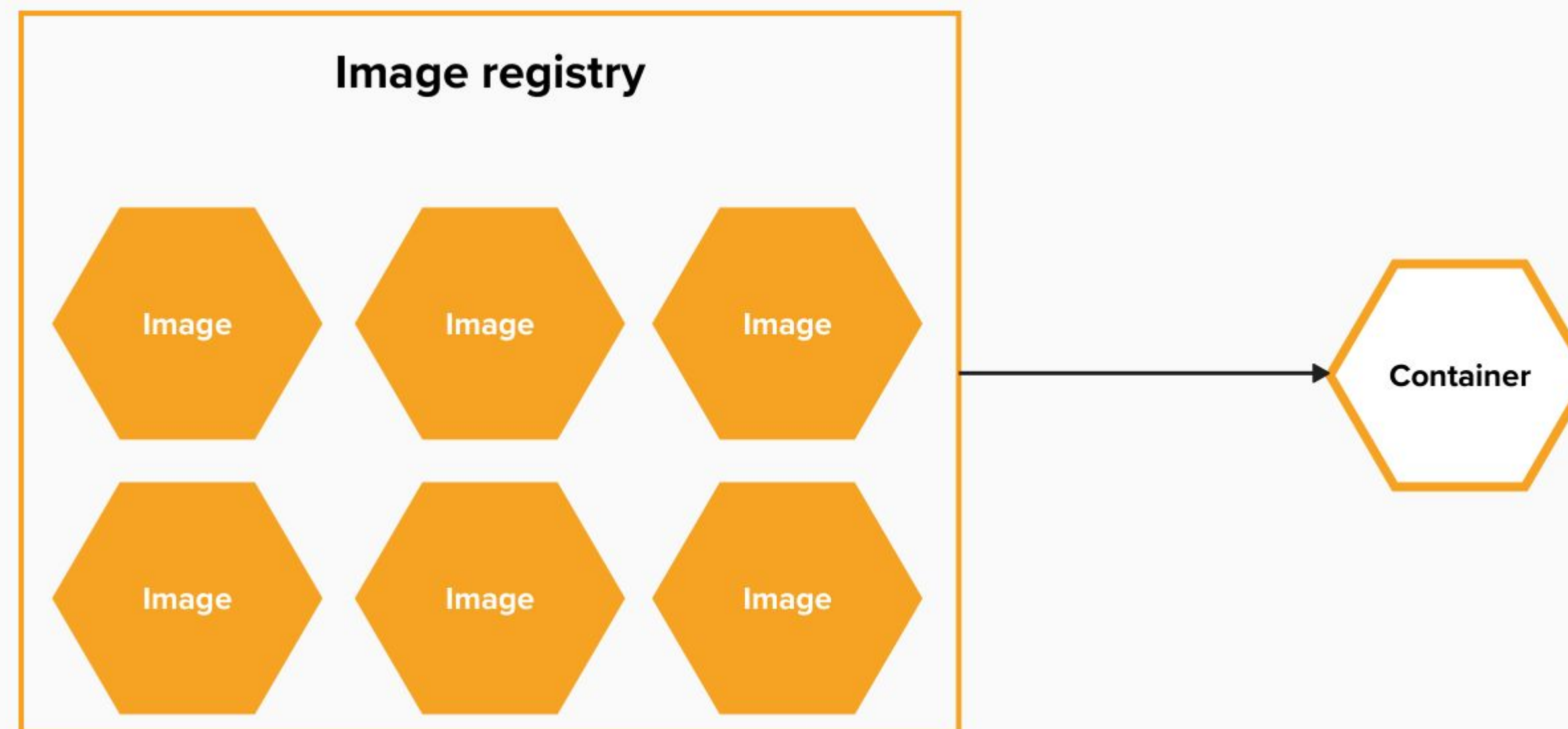
1. RECAP

containers are created from container images



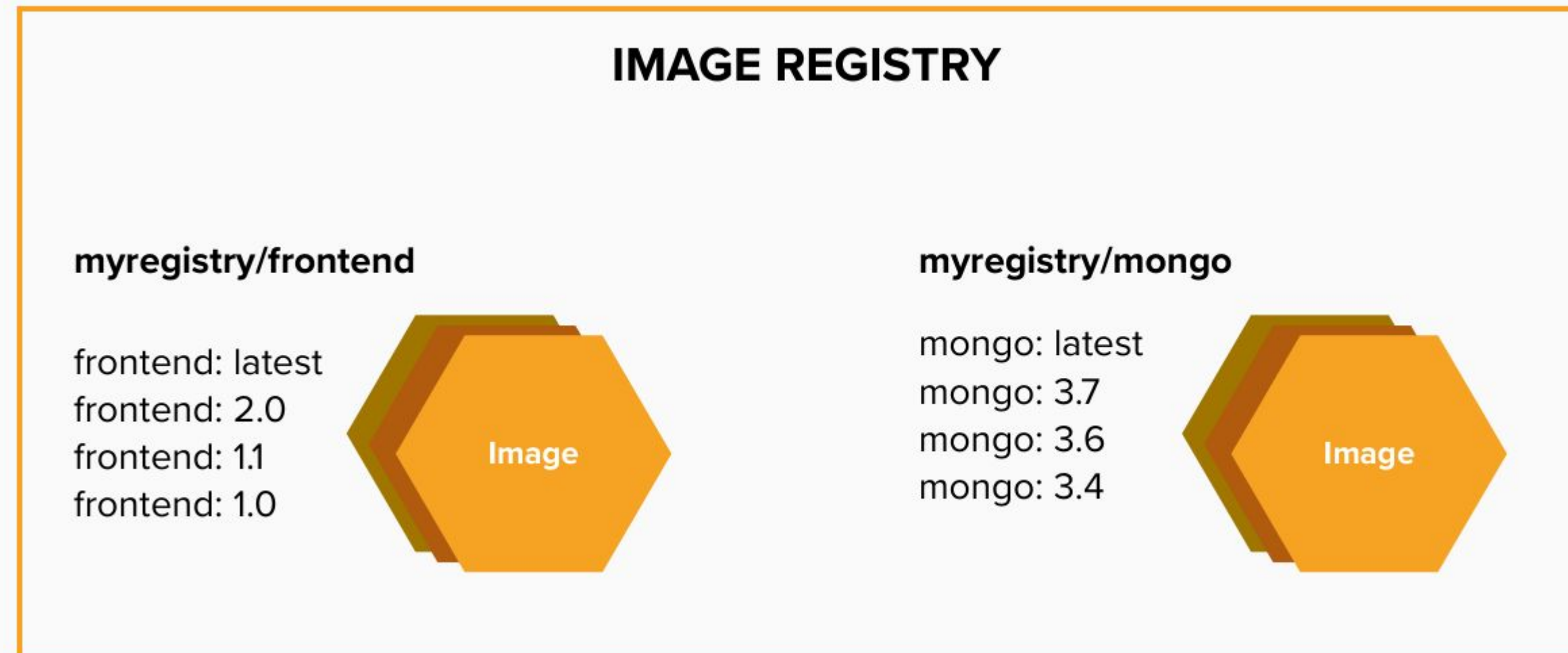
1. RECAP

container images are stored in an image registry



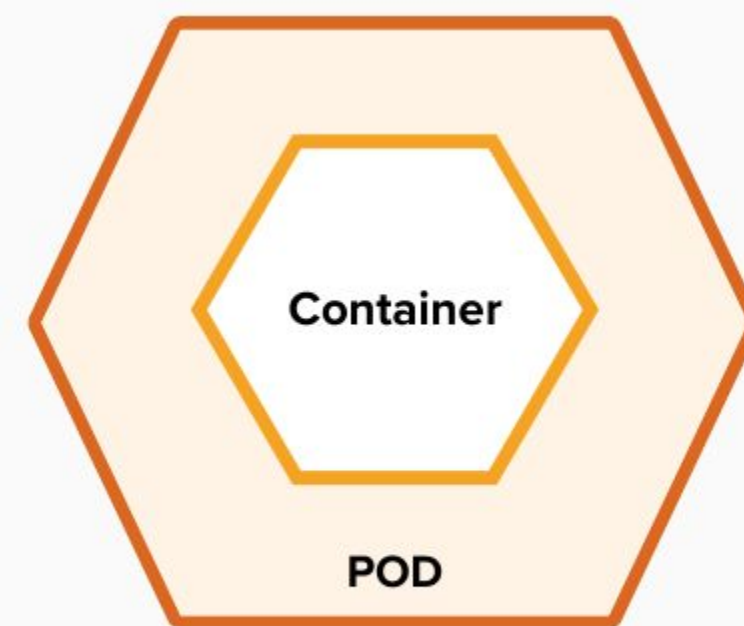
1. RECAP

an image repository contains all versions of an image in the image registry

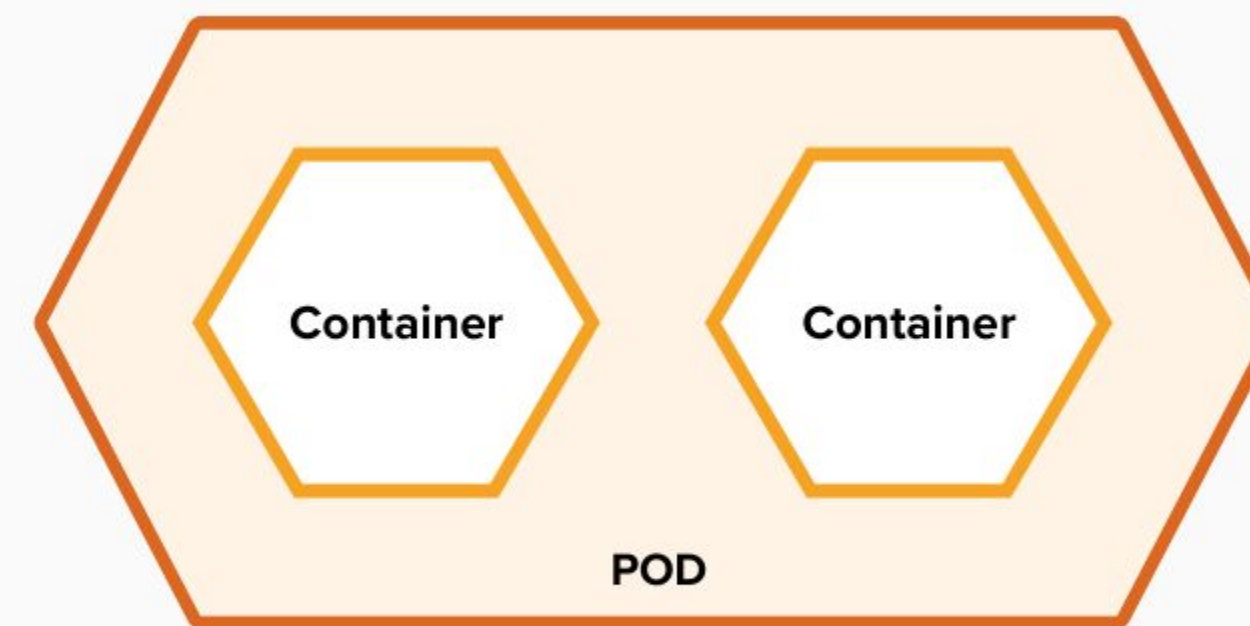


1. RECAP

containers are wrapped in pods which are units of deployment and management



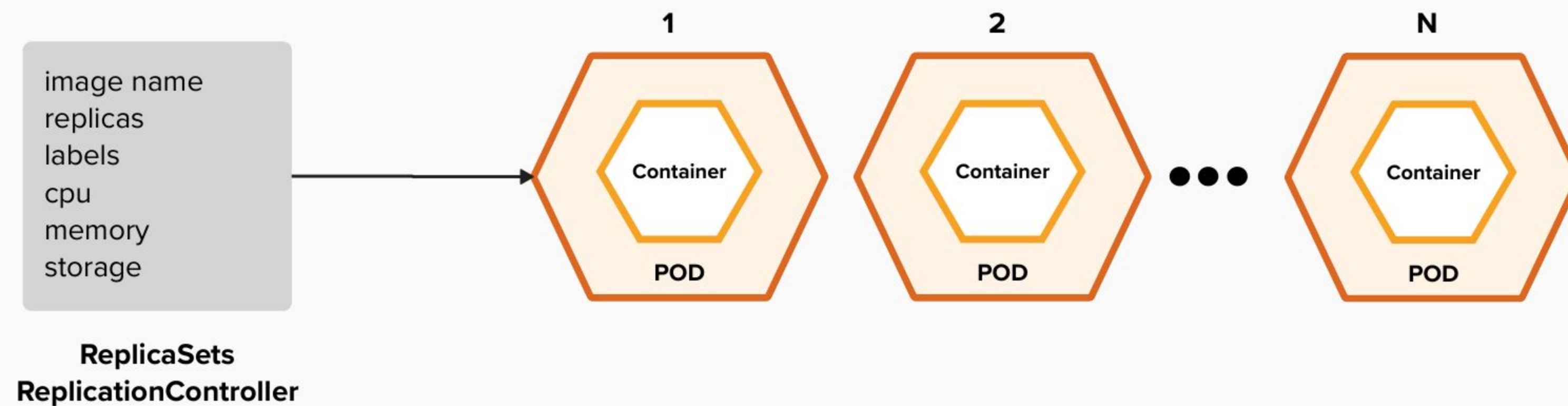
10.140.4.44



10.15.6.55

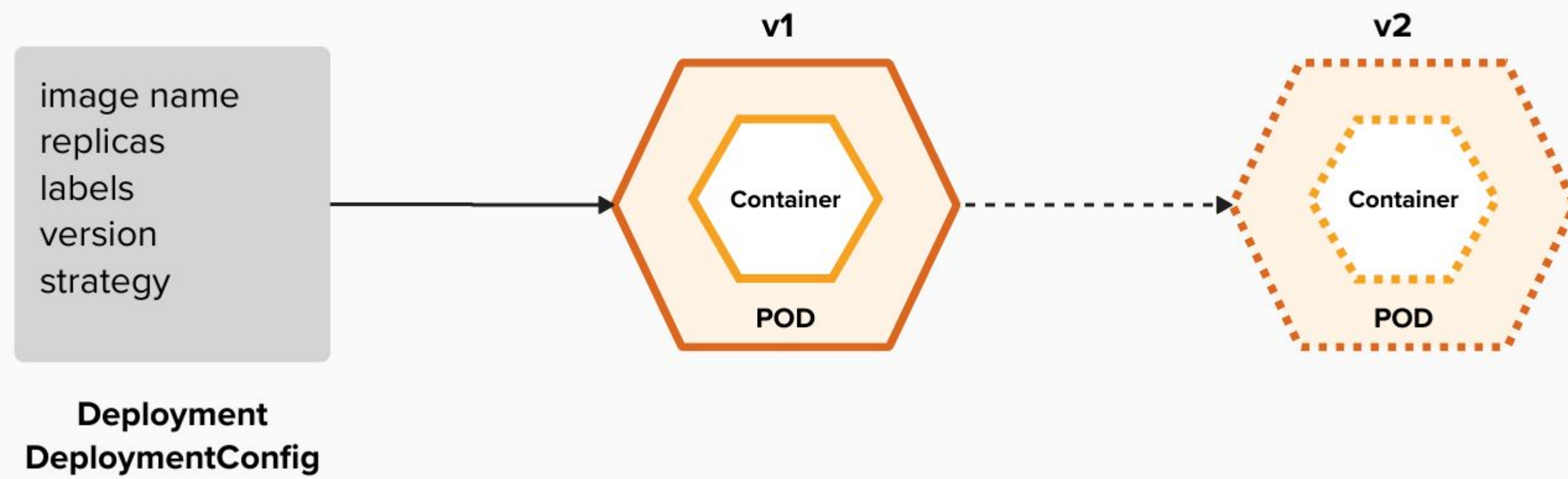
1. RECAP

ReplicationControllers & ReplicaSets ensure a specified number of pods are running at any given time



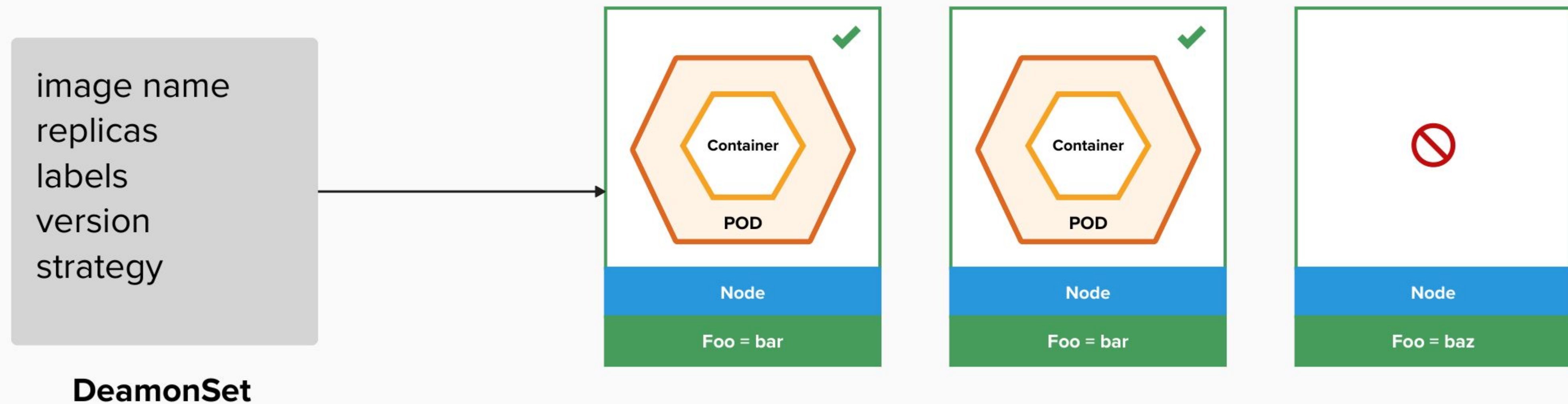
1. RECAP

Deployments and DeploymentConfigurations
define how to roll out new versions of Pods



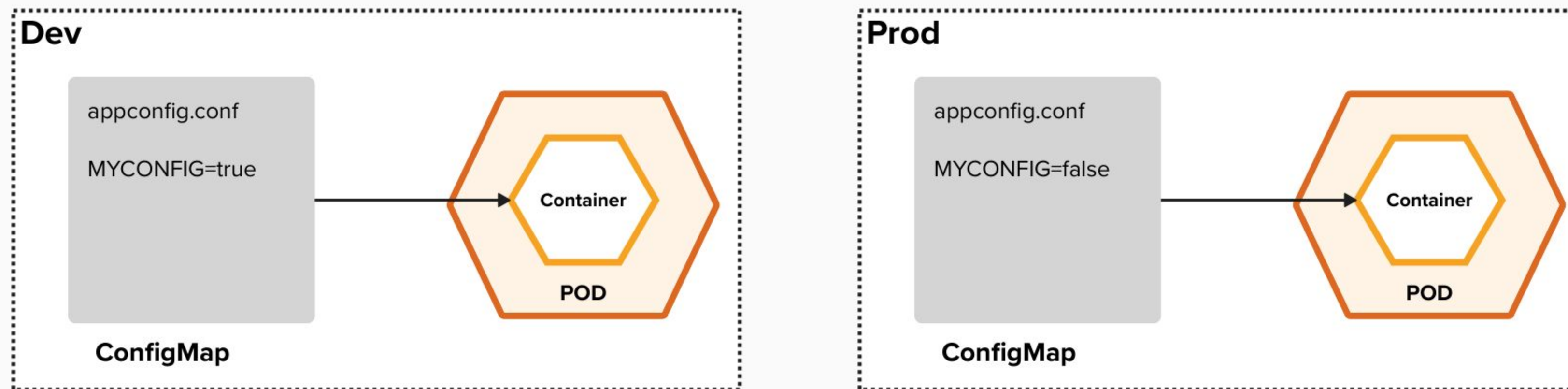
1. RECAP

a daemonset ensures that all (or some) nodes run
a copy of a pod



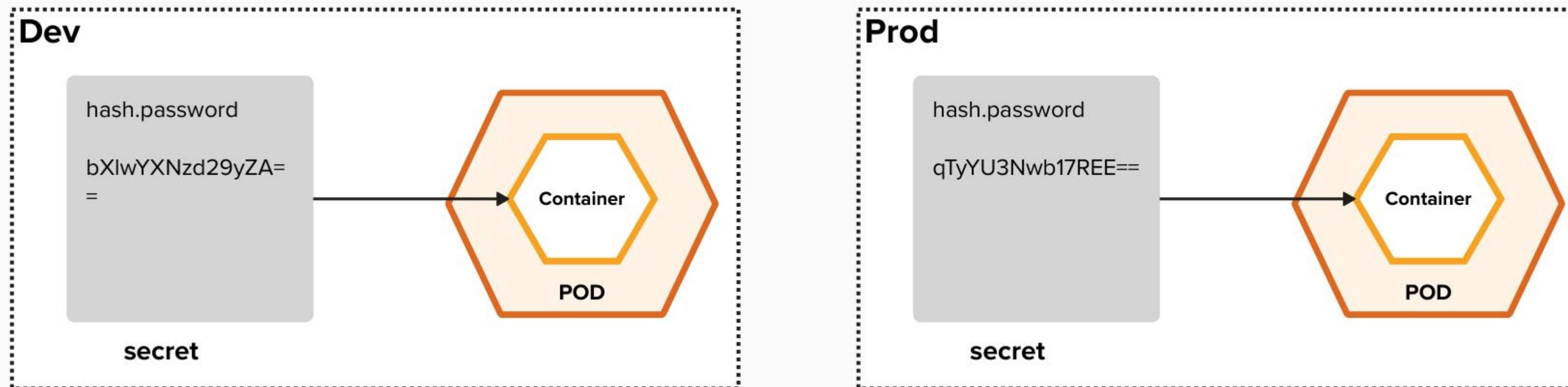
1. RECAP

configmaps allow you to decouple configuration artifacts from image content



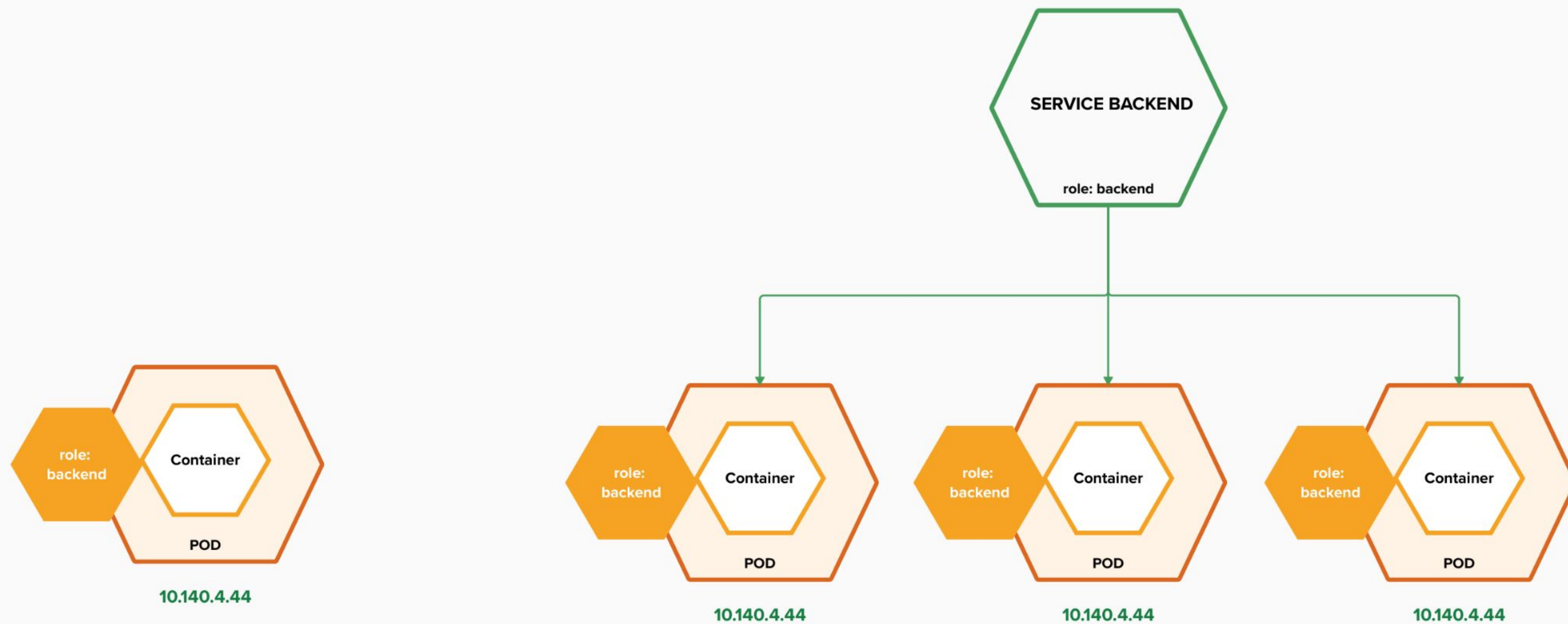
1. RECAP

secrets provide a mechanism to hold sensitive information such as passwords



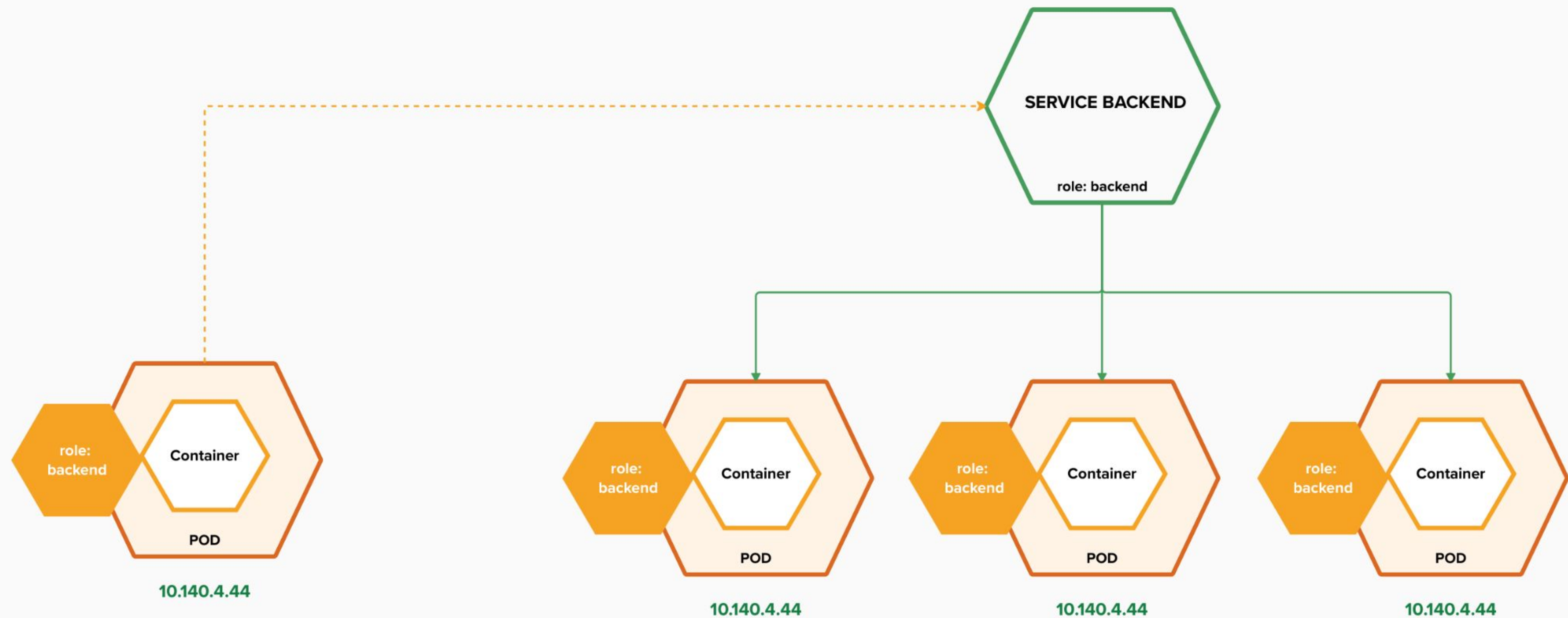
1. RECAP

services provide internal load-balancing and
service discovery across pods



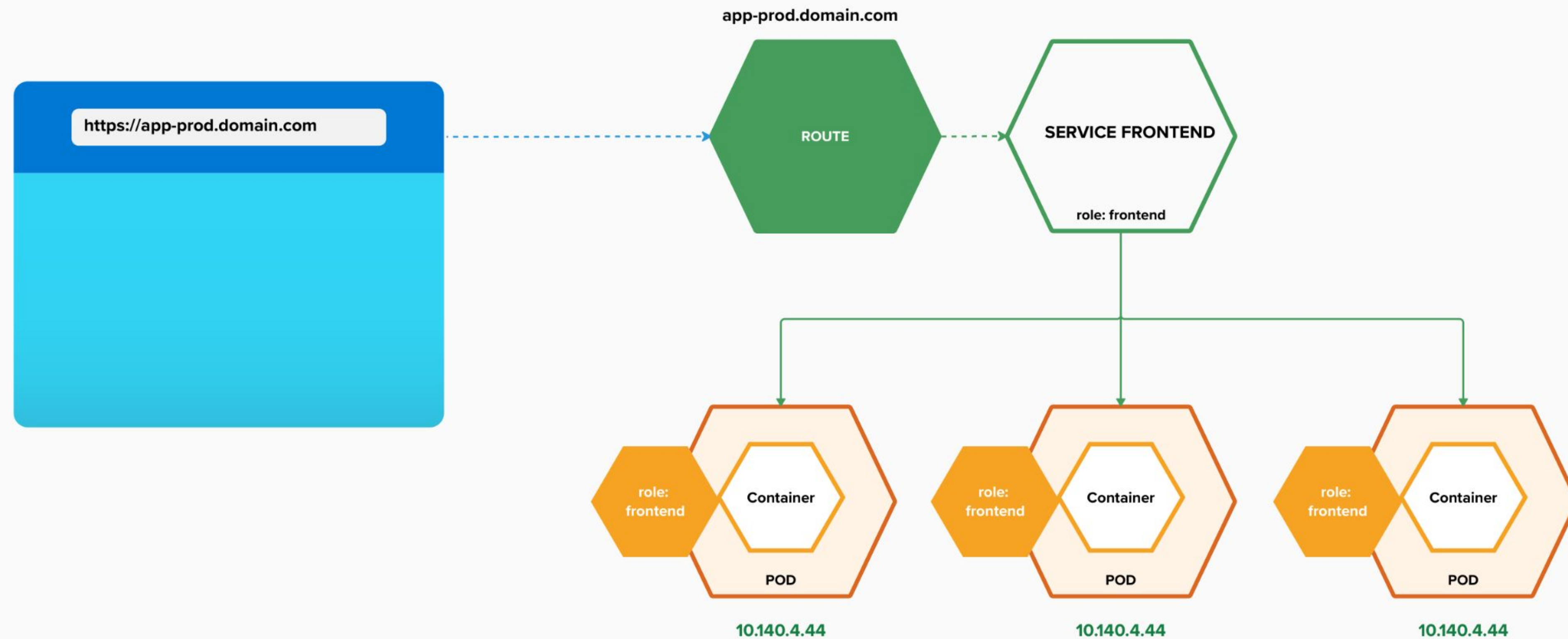
1. RECAP

apps can talk to each other via services



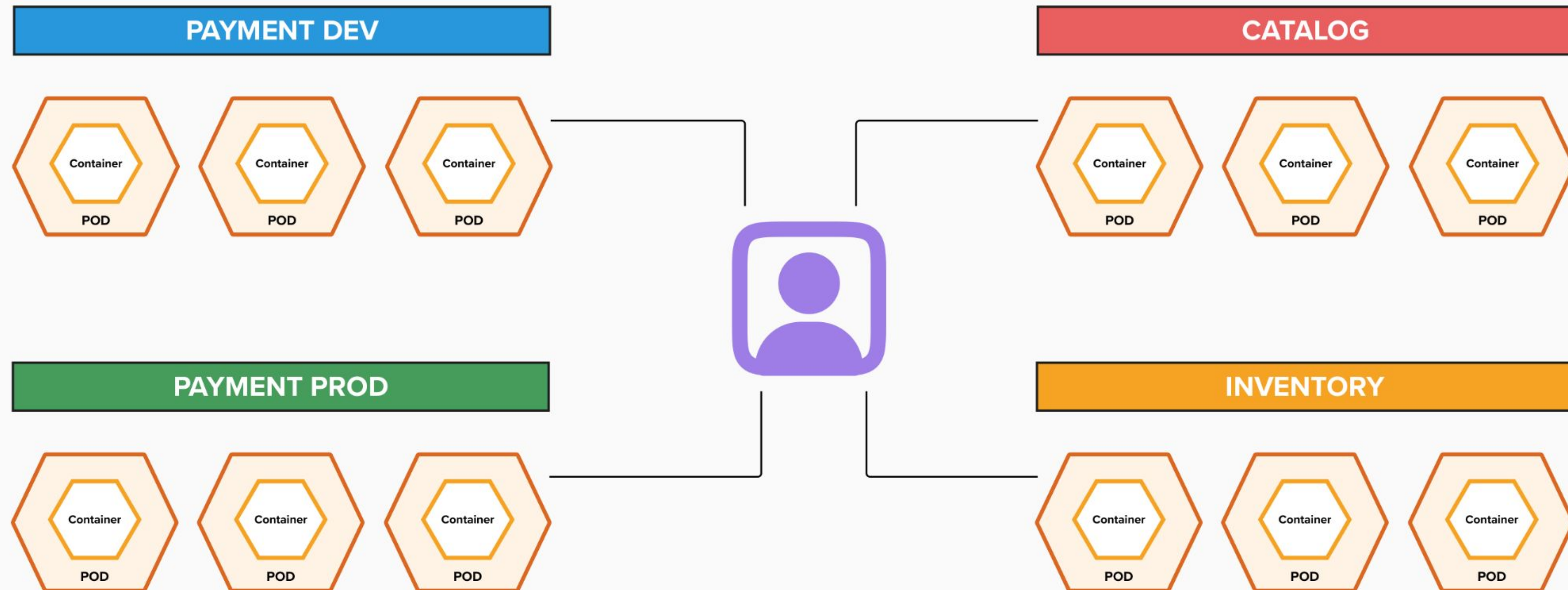
1. RECAP

routes make services accessible to clients outside the environment via real-world urls



1. RECAP

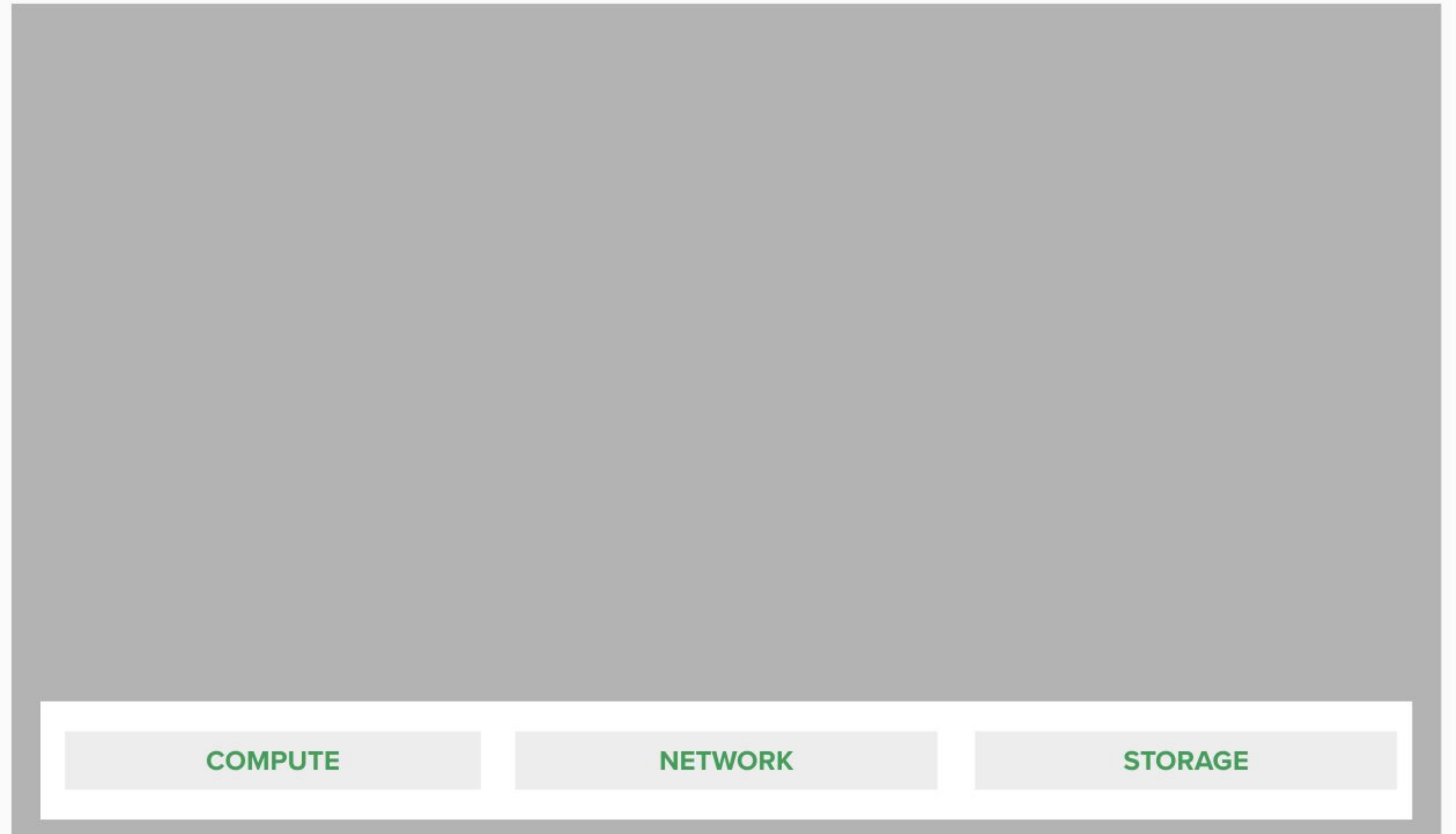
projects isolate apps across environments, teams,
groups and departments



OPENSIFT ARCHITECTURE

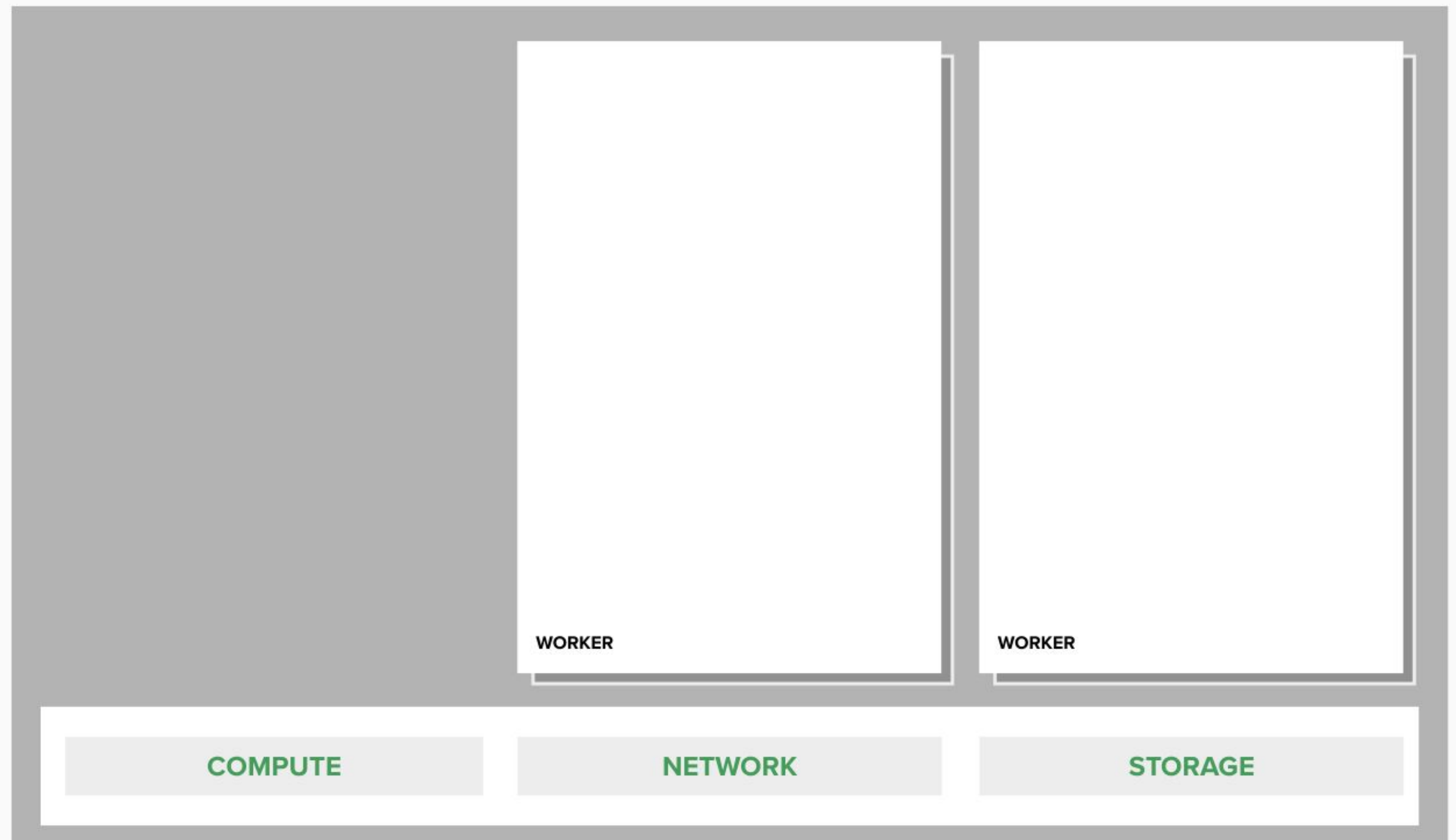
2. OPENSIFT ARCHITECTURE

your choice of infrastructure



2. OPENSIFT ARCHITECTURE

workers run workloads



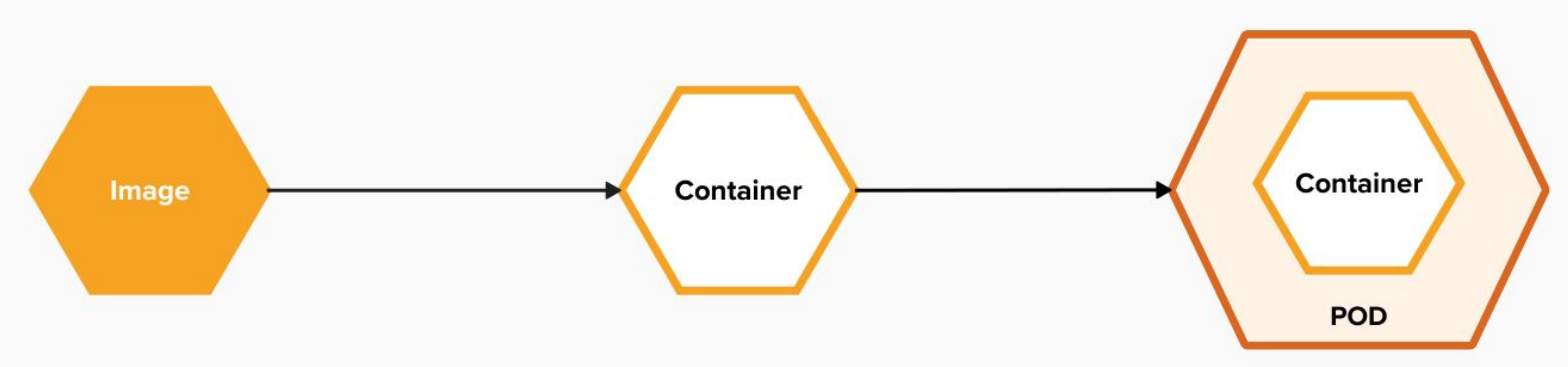
2. OPENSIFT ARCHITECTURE

masters are the control plane



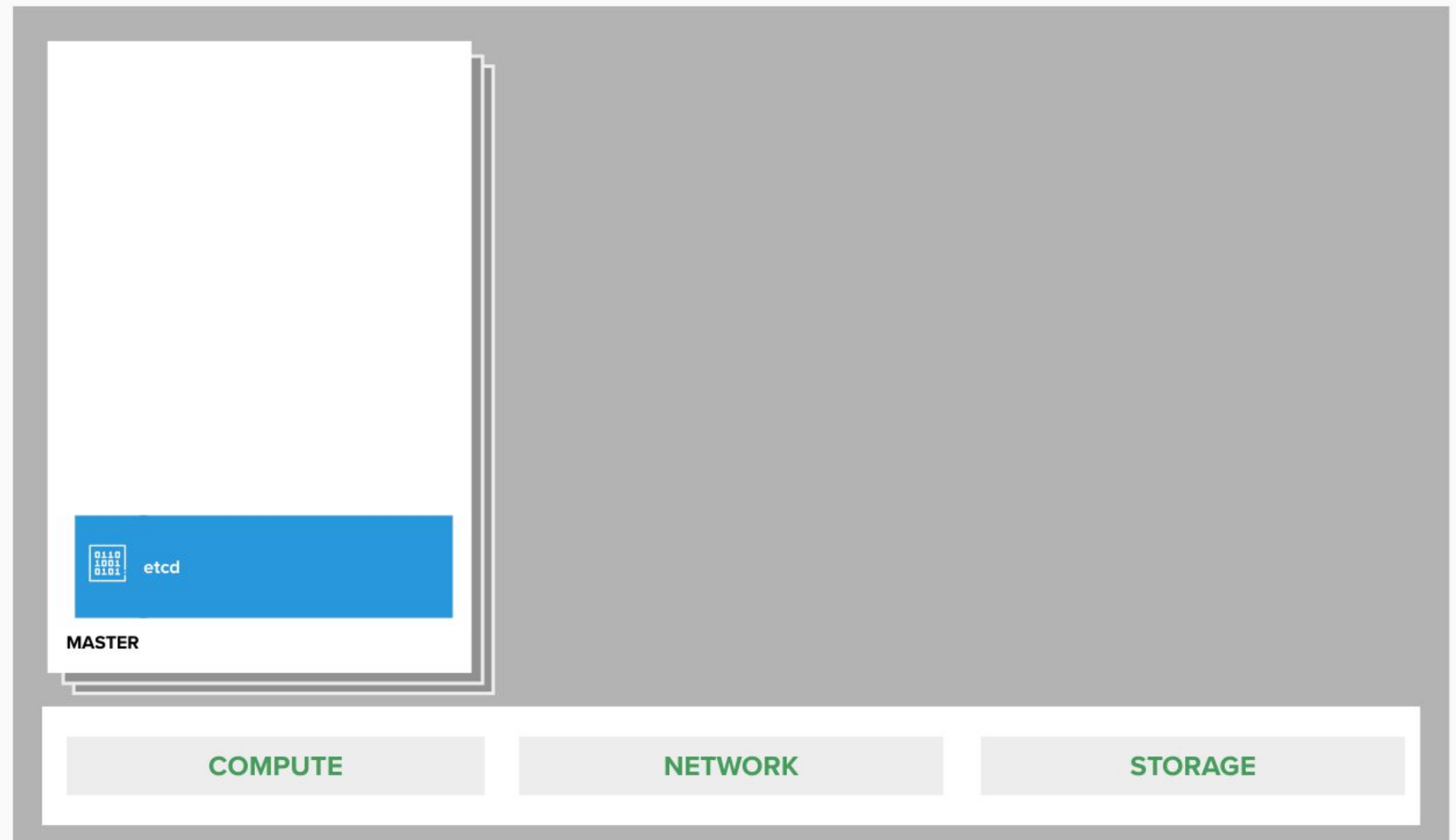
2. OPENSIFT ARCHITECTURE

everything runs in pods



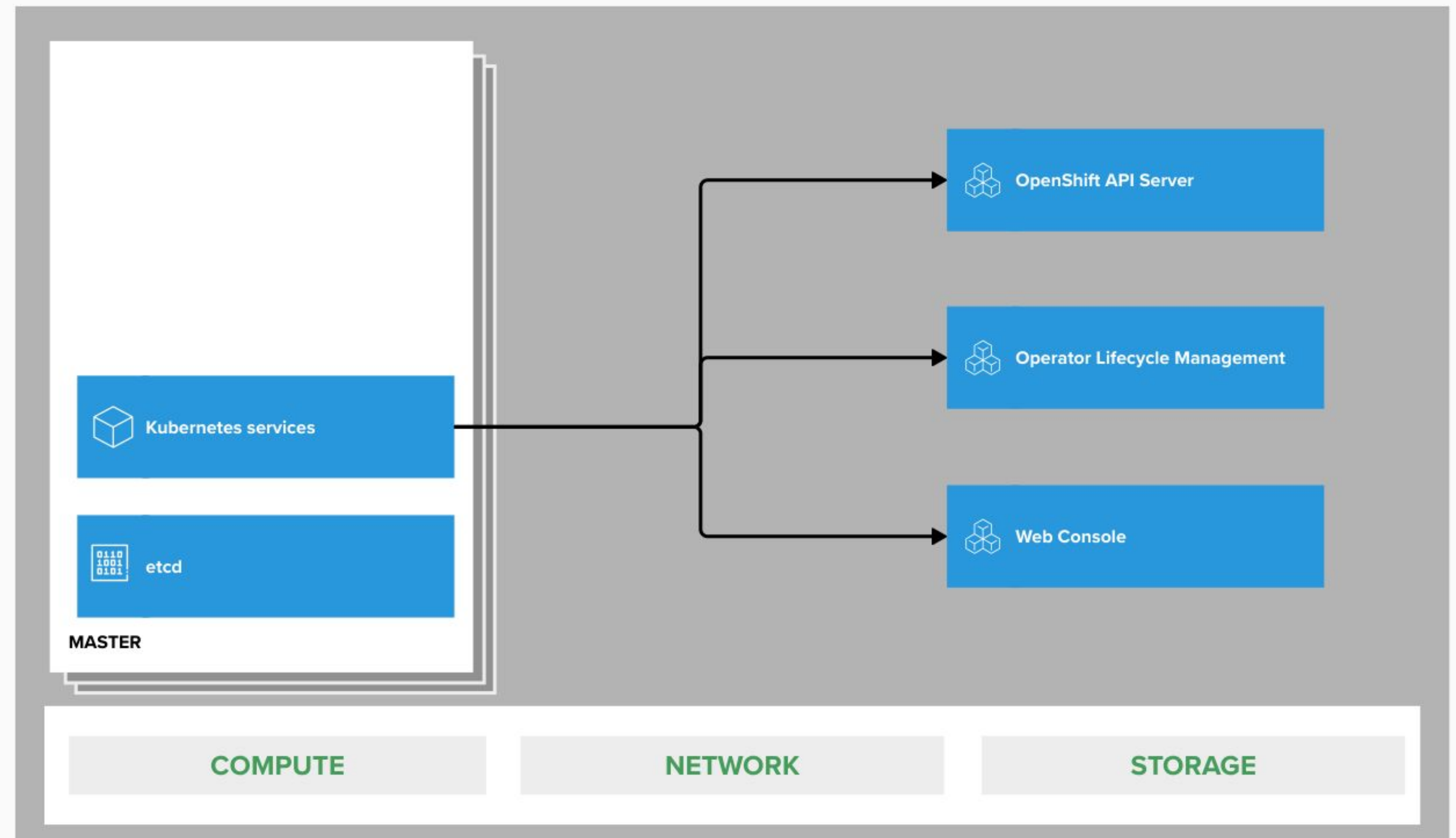
2. OPENSIFT ARCHITECTURE

state of everything



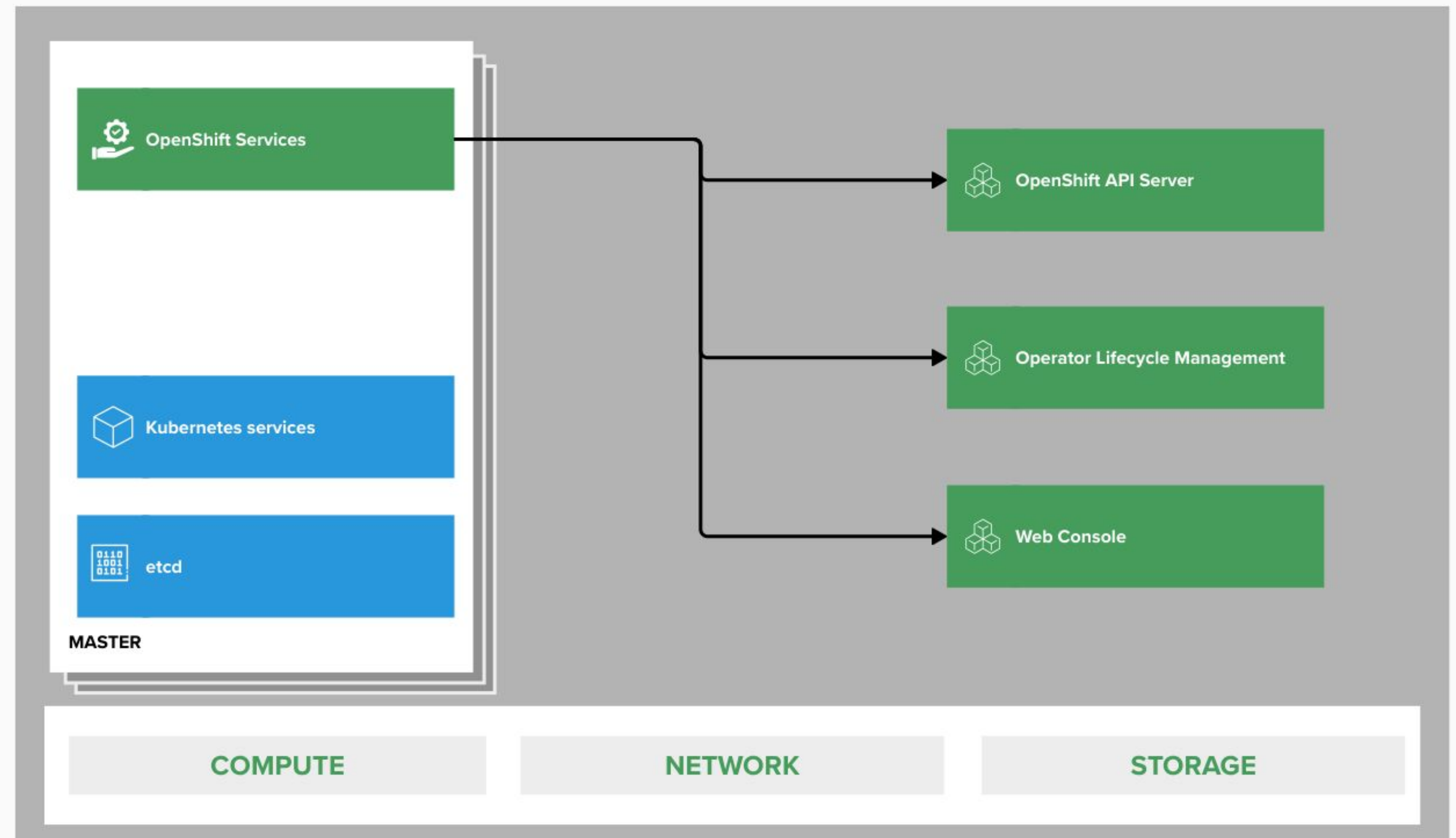
2. OPENSIFT ARCHITECTURE

core kubernetes components



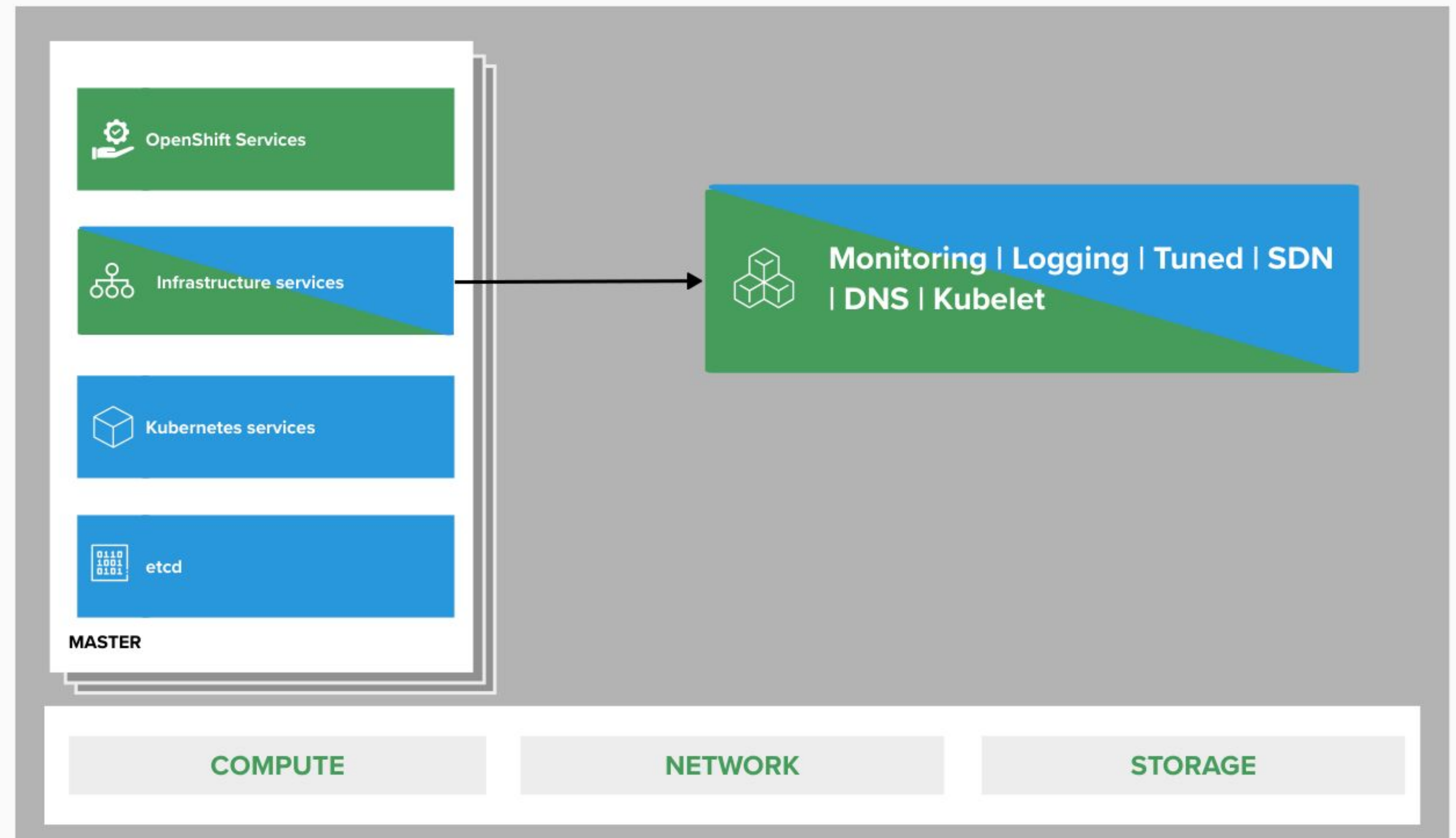
2. OPENSIFT ARCHITECTURE

core OpenShift components



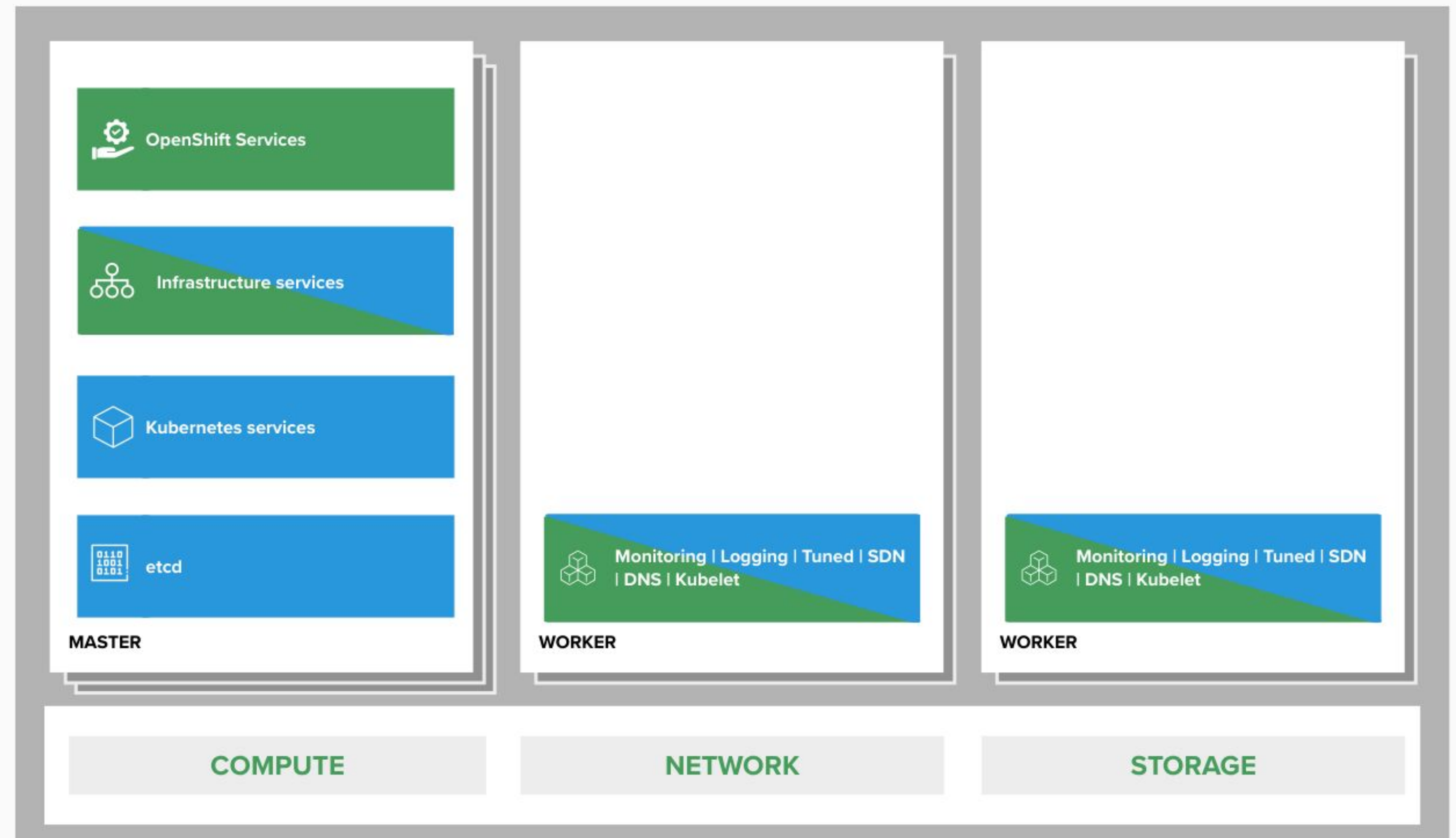
2. OPENSIFT ARCHITECTURE

internal and support infrastructure services



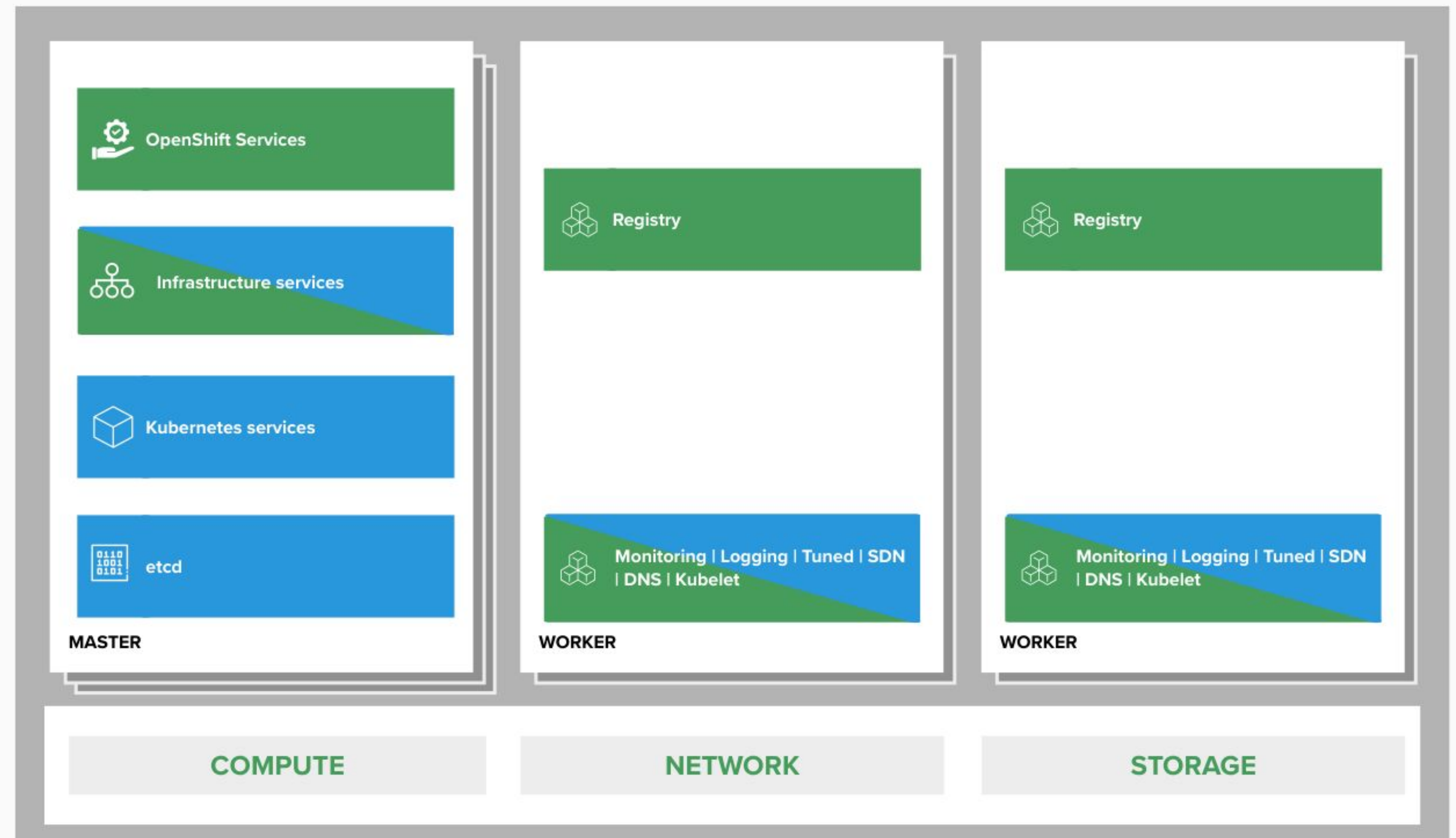
2. OPENSIFT ARCHITECTURE

run on all hosts



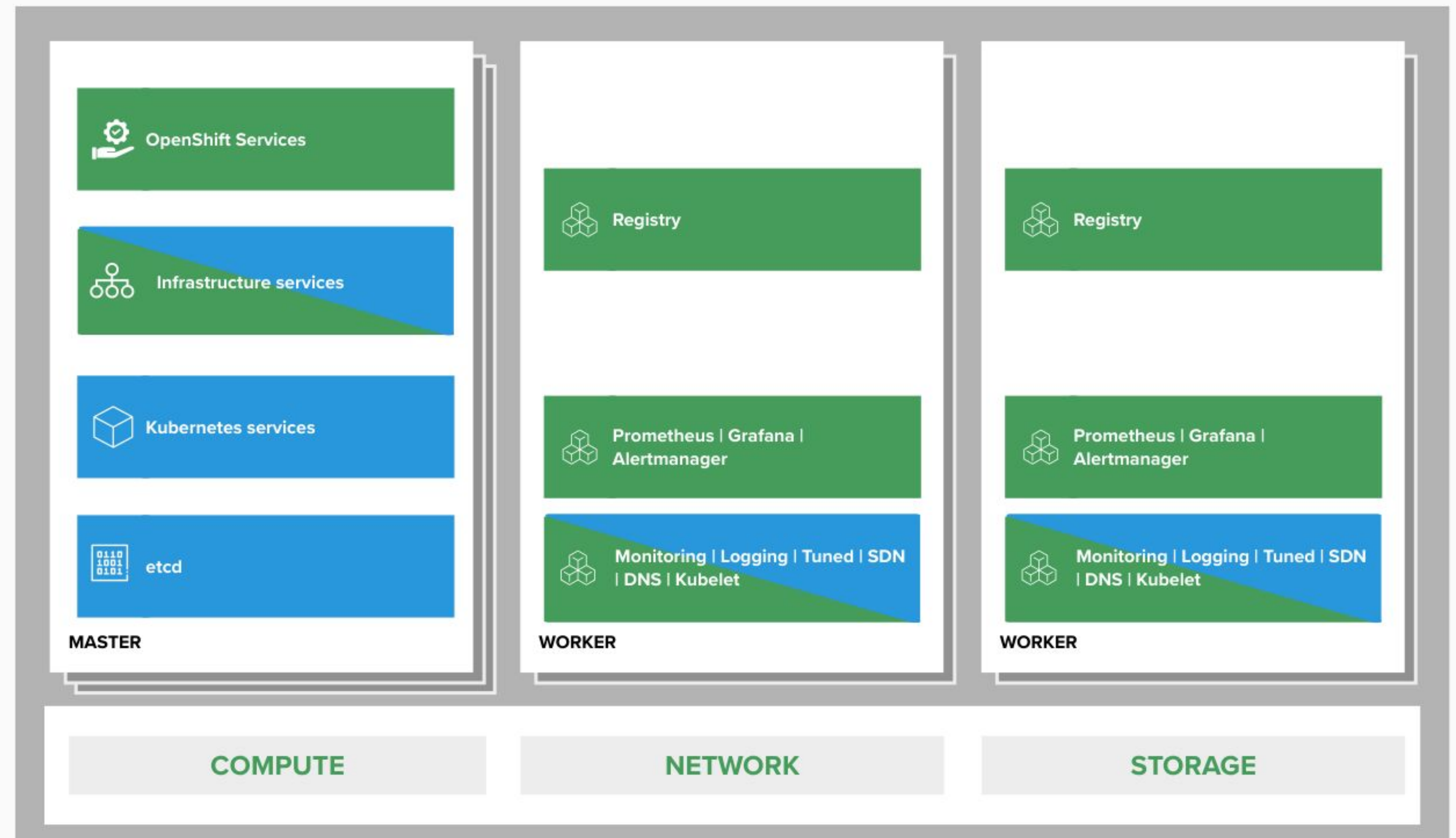
2. OPENSIFT ARCHITECTURE

integrated image registry



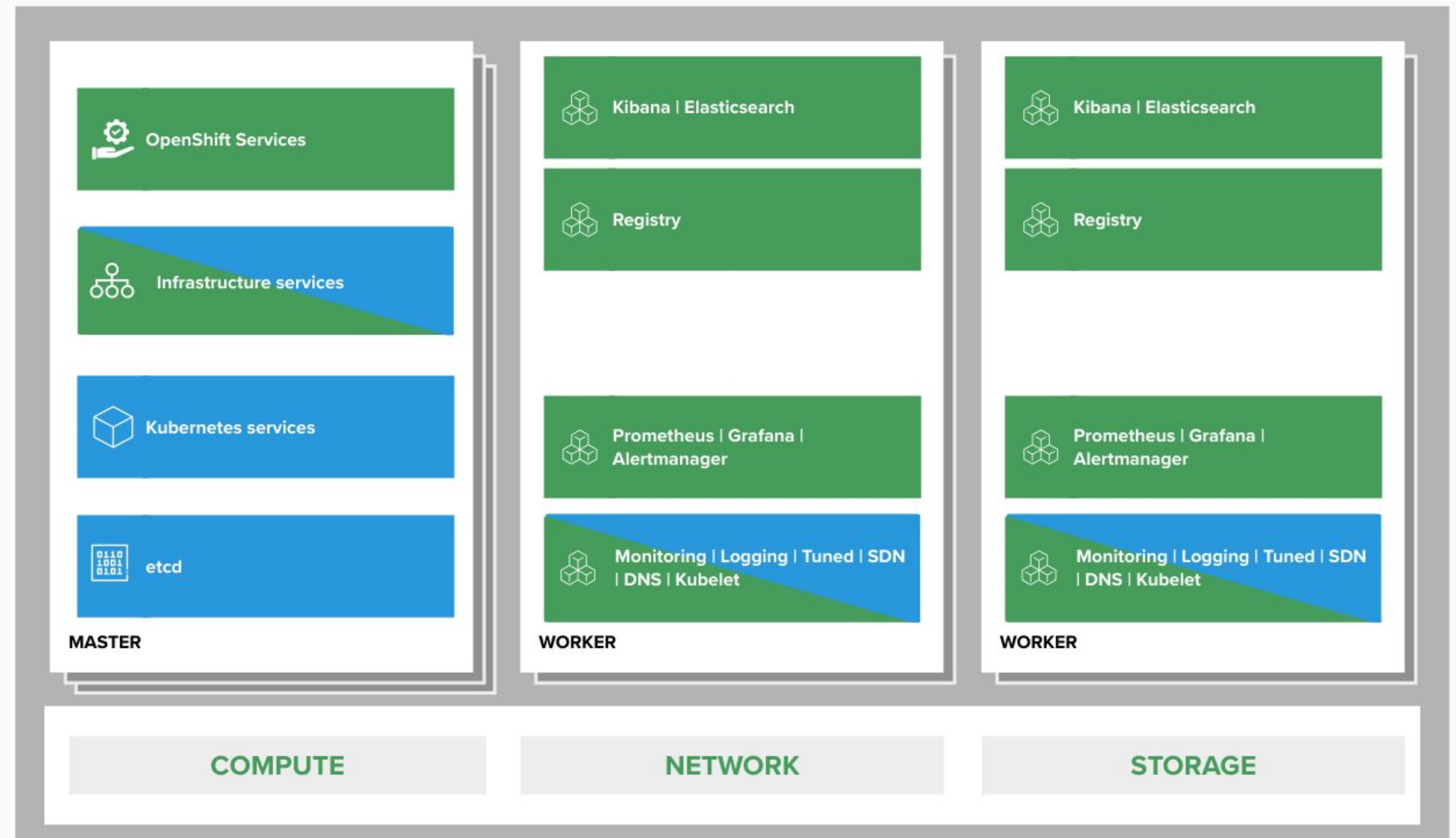
2. OPENSIFT ARCHITECTURE

cluster monitoring



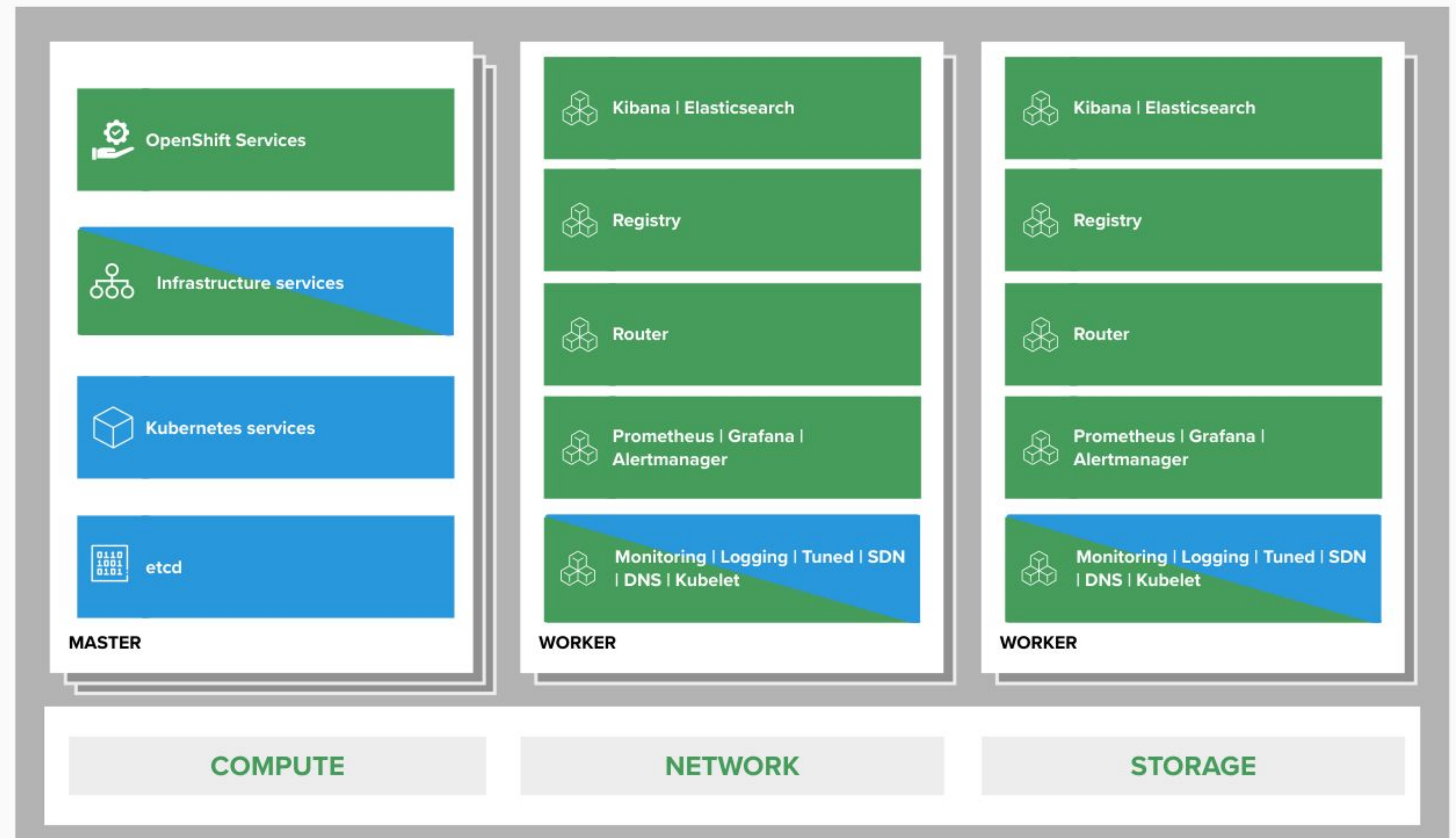
2. OPENSIFT ARCHITECTURE

log aggregation



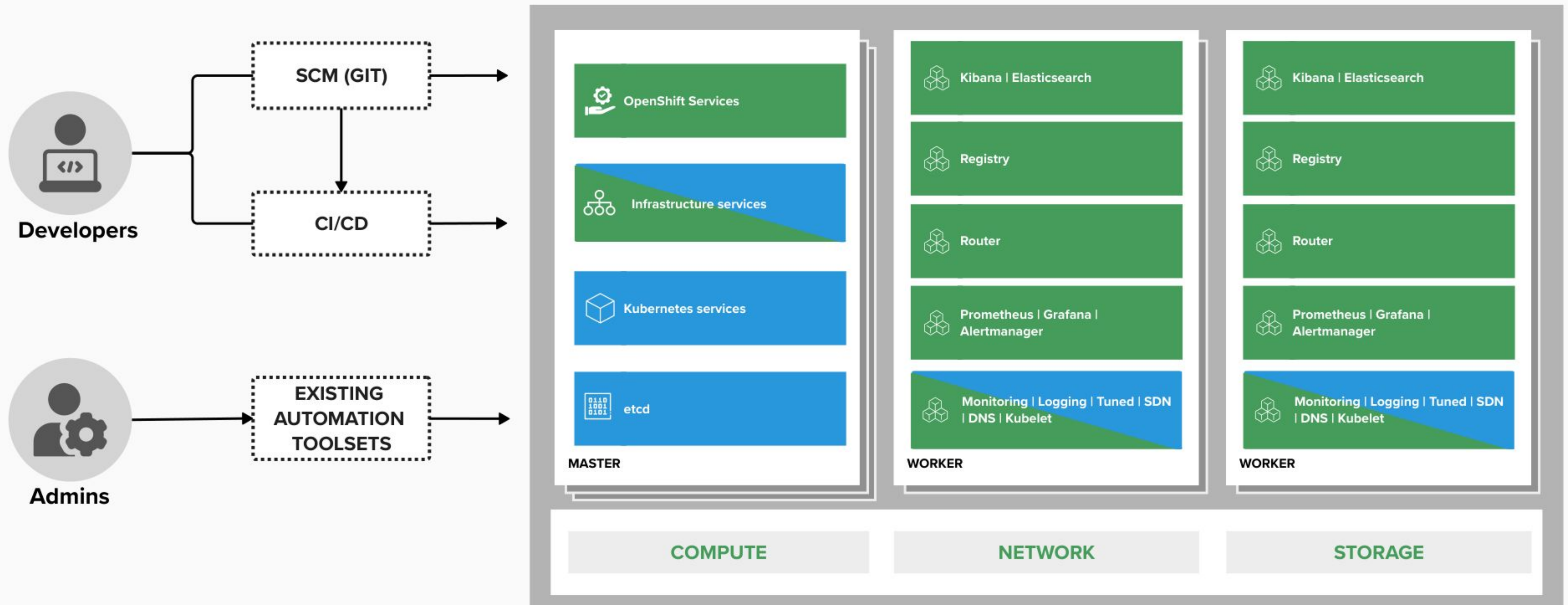
2. OPENSIFT ARCHITECTURE

integrated routing



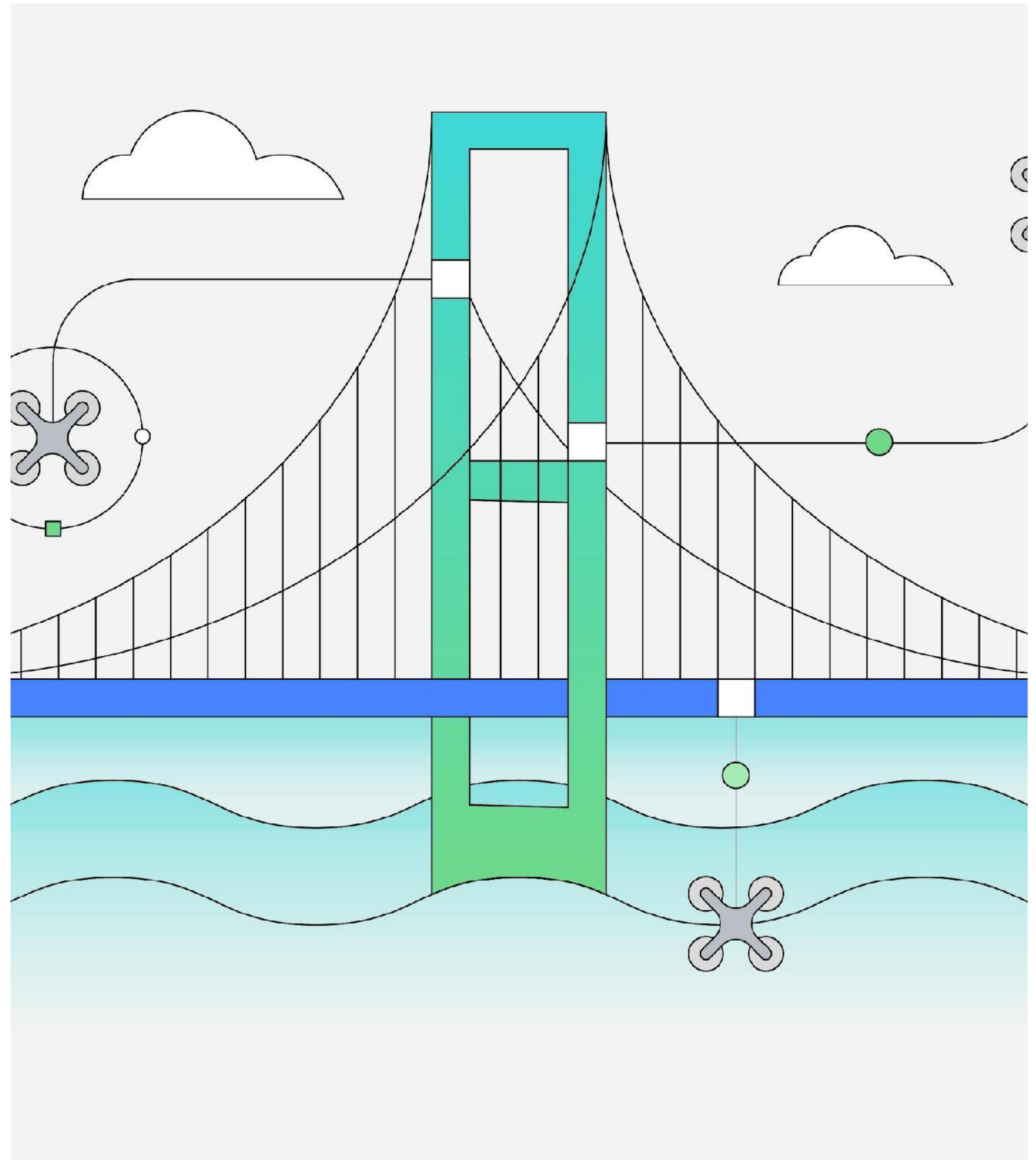
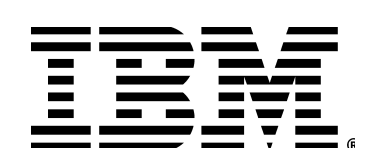
2. OPENSIFT ARCHITECTURE

dev and ops via web, cli, API, and IDE



3.

text



4.

text

5.

text

text